Rakamin Academy X kimia farma

Kimia Farma Business Performance Analytics

Kimia Farma - Big Data Analytics

Presented by
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Rakamin Academy

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Data Analyst Enthusiast

Management graduate from Gunadarma University with a strong interest in Data Analytics, Data Science, and Business Intelligence. Passionate about transforming data into actionable insights to support strategic decision-making. Detail-oriented, analytical, and eager to grow in data-driven roles.

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About Company

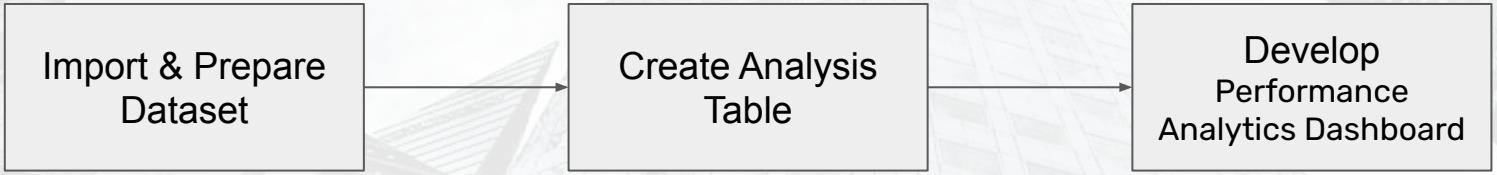
PT Kimia Farma Tbk is one of the largest and oldest pharmaceutical companies in Indonesia, established in 1817 during the Dutch colonial era. As a state-owned enterprise (SOE), Kimia Farma plays a strategic role in supporting national healthcare services.

The company operates across the entire pharmaceutical value chain, including manufacturing, distribution, retail pharmacy networks, clinical laboratories, and healthcare services. Kimia Farma produces a wide range of products such as ethical drugs, over-the-counter (OTC) medicines, herbal products, medical devices, and health supplements.

With thousands of pharmacy outlets and healthcare facilities across Indonesia, Kimia Farma is committed to improving public access to affordable and high-quality healthcare products and services. The company continuously innovates through digital transformation and data-driven strategies to strengthen operational efficiency and customer experience.

Project Overview

In this project, I conducted a business performance analysis for Kimia Farma using data from 2020 to 2023. The workflow was structured into three key phases: dataset importation and preparation, creation of an analysis table, and development of a Performance Analytics Dashboard to present actionable insights.



Github

Tools

Google BigQuery

For data preparation and data analysis.

Looker Studio

Support data visualization and interactive dashboard creation

Project Framework

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1. Business Understanding
2. Data Collection
3. Data Preparation
4. Data Analysis
5. Data Visualization
6. Insight & Recommendation

Business Understanding

The objective of this project is to analyze Kimia Farma's business performance from 2020 to 2023 to evaluate financial trends, regional performance, product contribution, and customer satisfaction.

This analysis focuses on identifying key performance indicators such as **net sales, profit, total transactions, and customer ratings** in order to understand growth patterns, detect performance gaps across provinces, and uncover opportunities for business optimization and strategic decision-making.

Data Collection

This project involves **four** primary data categories: **transactions**, **inventory**, **Kimia Farma branch information**, and **products.**

kf_final_transaction.csv

Sales transaction dataset with fields:
transaction_id, product_id, branch_id, customer_name, date, price, discount_percentage, rating.

kf_inventory.csv

Inventory dataset with fields:
inventory_id, branch_id, product_id, product_name, opname_stock.

Data Collection

This project involves **four** primary data categories: **transactions**, **inventory**, **Kimia Farma branch information**, and **products.**

kf_kantor_cabang.csv

Branch dataset with fields:
branch_id, branch_category, branch_name, kota, provinsi, rating.

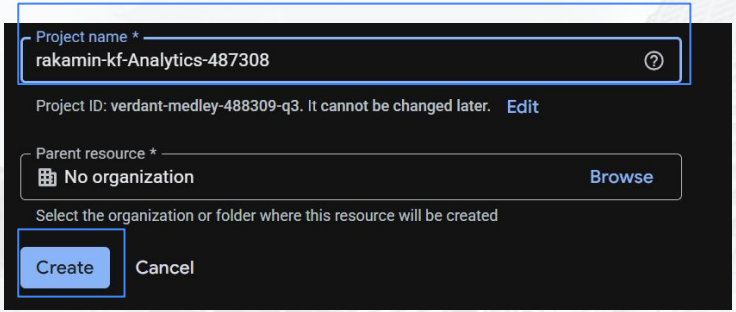
kf_product.csv

Product dataset with fields:
product_id, product_name, product_category, price.

Data Collection

The first step was importing the dataset into Google BigQuery. Before that, a new project was created, followed by setting up a dataset and creating tables by importing the four previously provided csv files.

1



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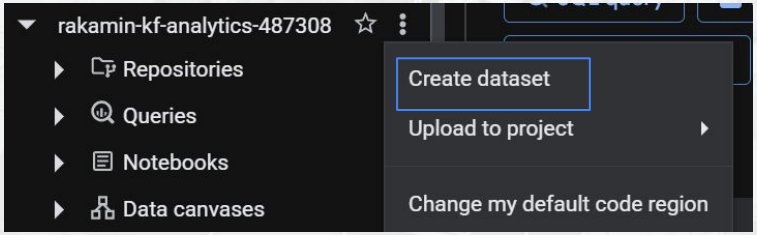

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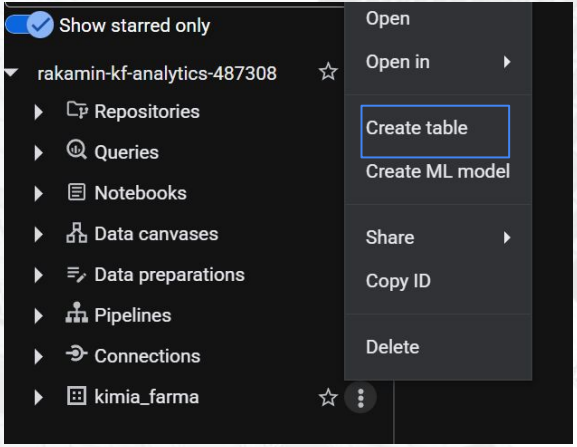


Data Collection

kf_final_transa... | Query | Create conve

Schema | Details | Preview | Table Explorer | Preview

Filter Enter property name or value

Field name	Type	Mode
transaction_id	STRING	NULLABLE
date	DATE	NULLABLE
branch_id	INTEGER	NULLABLE
customer_name	STRING	NULLABLE
product_id	STRING	NULLABLE
price	INTEGER	NULLABLE
discount_percentage	FLOAT	NULLABLE
rating	FLOAT	NULLABLE

kf_kantor_cab... | Query | Ope

Schema | Details | Preview | Table Explorer | Prev

Filter Enter property name or value

Field name	Type	Mode
branch_id	INTEGER	NULLABLE
branch_category	STRING	NULLABLE
branch_name	STRING	NULLABLE
kota	STRING	NULLABLE
provinsi	STRING	NULLABLE
rating	FLOAT	NULLABLE

kf_product | Query | Open in

Schema | Details | Preview | Table Explorer | Previ

Filter Enter property name or value

Field name	Type	Mode
product_id	STRING	NULLABLE
product_name	STRING	NULLABLE
product_category	STRING	NULLABLE
price	INTEGER	NULLABLE

kf_inventory | Query | Open i

Schema | Details | Preview | Table Explorer | P

Filter Enter property name or value

Field name	Type	Mode
Inventory_ID	STRING	NULLABLE
branch_id	INTEGER	NULLABLE
product_id	STRING	NULLABLE
product_name	STRING	NULLABLE
opname_stock	INTEGER	NULLABLE

Data Preparation

Following the data import process, data preparation was conducted by verifying data types and ensuring consistency. As the dataset was already clean and properly structured, the analysis phase could be carried out without additional preprocessing steps.

kf_final_transa... Query Create conve

Schema Details Preview Table Explorer Preview

Filter Enter property name or value

Field name	Type	Mode
transaction_id	STRING	NULLABLE
date	DATE	NULLABLE
branch_id	INTEGER	NULLABLE
customer_name	STRING	NULLABLE
product_id	STRING	NULLABLE
price	INTEGER	NULLABLE
discount_percentage	FLOAT	NULLABLE
rating	FLOAT	NULLABLE

kf_kantor_cab... Query Ope

Schema Details Preview Table Explorer Pre

Filter Enter property name or value

Field name	Type	Mode
branch_id	INTEGER	NULLABLE
branch_category	STRING	NULLABLE
branch_name	STRING	NULLABLE
kota	STRING	NULLABLE
provinsi	STRING	NULLABLE
rating	FLOAT	NULLABLE

kf_product Query Open in

Schema Details Preview Table Explorer Previ

Filter Enter property name or value

Field name	Type	Mode
product_id	STRING	NULLABLE
product_name	STRING	NULLABLE
product_category	STRING	NULLABLE
price	INTEGER	NULLABLE

kf_inventory Query Open

Schema Details Preview Table Explorer P

Filter Enter property name or value

Field name	Type	Mode
Inventory_ID	STRING	NULLABLE
branch_id	INTEGER	NULLABLE
product_id	STRING	NULLABLE
product_name	STRING	NULLABLE
opname_stock	INTEGER	NULLABLE

Data Analysis

The next step involved creating an analysis table containing aggregated results from the previously prepared tables. Subsequently, the data analysis process was conducted based on this analysis table.

```sql
-- Create a new table to store all of the relevant attributes
CREATE OR REPLACE TABLE
`rakamin-kf-analytics-487308.kimia_farma.analysis_table` AS
SELECT
    t.transaction_id,
    t.date,
    t.branch_id,
    kc.branch_name,
    kc.kota,
    kc.provinsi,
    kc.rating AS rating_cabang,
    t.customer_name,
    t.product_id,
    p.product_name,
    t.price AS actual_price,
    t.discount_percentage,

    CASE
        WHEN t.price <= 50000 THEN 0.10
        WHEN t.price > 50000 AND t.price <= 100000 THEN 0.15
        WHEN t.price > 100000 AND t.price <= 300000 THEN 0.20
        WHEN t.price > 300000 AND t.price <= 500000 THEN 0.25
        ELSE 0.30
    END AS gross_profit_percentage,

    t.price * (1 - t.discount_percentage) AS net_sales,

    t.price * (1 - t.discount_percentage) *
    CASE
        WHEN t.price <= 50000 THEN 0.10
        WHEN t.price > 50000 AND t.price <= 100000 THEN 0.15
        WHEN t.price > 100000 AND t.price <= 300000 THEN 0.20
        WHEN t.price > 300000 AND t.price <= 500000 THEN 0.25
        ELSE 0.30
    END AS net_profit,

    t.rating AS rating_transaksi

FROM `rakamin-kf-analytics-487308.kimia_farma.kf_final_transaction` t
LEFT JOIN `rakamin-kf-analytics-487308.kimia_farma.kf_product` p
    ON t.product_id = p.product_id
LEFT JOIN `rakamin-kf-analytics-487308.kimia_farma.kf_kantor_cabang` kc
    ON t.branch_id = kc.branch_id;
```

analysis_table Query Create conversation

Schema Details Preview Table Explorer Preview

Filter Enter property name or value

Field name	Type	Mode
transaction_id	STRING	NULLABLE
date	DATE	NULLABLE
branch_id	INTEGER	NULLABLE
branch_name	STRING	NULLABLE
kota	STRING	NULLABLE
provinsi	STRING	NULLABLE
rating_cabang	FLOAT	NULLABLE
customer_name	STRING	NULLABLE
product_id	STRING	NULLABLE
product_name	STRING	NULLABLE
actual_price	INTEGER	NULLABLE
discount_percentage	FLOAT	NULLABLE
gross_profit_percentage	FLOAT	NULLABLE
net_sales	FLOAT	NULLABLE
net_profit	FLOAT	NULLABLE
rating_transaksi	FLOAT	NULLABLE

Data Analysis

The next step involved creating an analysis table containing aggregated results from the previously prepared tables. Subsequently, the data analysis process was conducted based on this analysis table.

Net Profit
Rp91.2 M

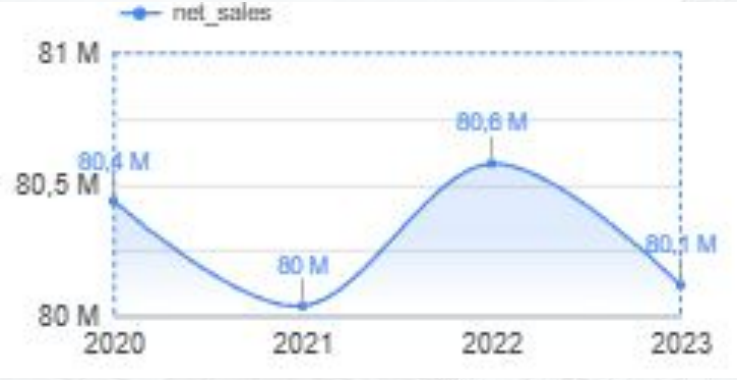
Net Sales
Rp321.2 M

Total Transaction
672.5 rb

Transaction Rating
4

Data Analysis

Net Sales Stable yet Fluctuating



Net sales recorded **80.4M in 2020**, declined slightly to **80.0M in 2021**, indicating a temporary slowdown, then rebounded to a peak of **80.6M in 2022**, reflecting strong recovery and improved business performance, before experiencing a minor decrease to **80.1M in 2023**; overall, the trend demonstrates **stable yet fluctuating performance**, with **2022 standing out as the strongest growth year** following the dip in 2021.

Data Analysis

Profit Stable with Minor Fluctuating

Profit reached **22.8M in 2020**, declined slightly to **22.7M in 2021**, then increased to a peak of **22.9M in 2022**, reflecting improved profitability, before easing back to **22.8M in 2023**; overall, the trend indicates **relatively stable performance with minor fluctuations**, with **2022 marking the highest profit level** during the observed period.

Data Analysis

West Java Emerges as the Top Revenue Contributor

Top 10 Provincial Branches by Net Sales

1.	Jawa Barat	94,9 M
2.	Sumatera Utara	23 M
3.	Jawa Tengah	22,2 M
4.	Jawa Timur	16,6 M
5.	Sulawesi Utara	15,9 M
6.	Sumatera Barat	15,3 M
7.	Aceh	10,5 M
8.	Nusa Tenggara Barat	10 M
9.	Kalimantan Timur	9,7 M
10.	Riau	9,4 M

West Java ranked first with **94.9M**, significantly outperforming other provinces and contributing the largest share of total sales. Sumatera Utara followed with **23M**, while Jawa Tengah recorded **22.2M**, placing third. Jawa Timur and Sulawesi Utara generated **16.6M** and **15.9M**, respectively, showing moderate performance compared to the top three. The remaining provinces—Sumatera Barat (**15.3M**), Aceh (**10.5M**), Nusa Tenggara Barat (**10M**), Kalimantan Timur (**9.7M**), and Riau (**9.4M**)—contributed smaller but still meaningful portions of total sales; overall, the distribution indicates a **strong concentration of sales in West Java**, with a noticeable gap between the leading province and the others.

Data Analysis

Transaction Activity Concentrated in West Java

Top 10 Provincial Branches by Total Transactions		
1.	Jawa Barat	198,7 rb
2.	Sumatera Utara	48,2 rb
3.	Jawa Tengah	46,5 rb
4.	Jawa Timur	34,8 rb
5.	Sulawesi Utara	33,3 rb
6.	Sumatera Barat	32 rb
7.	Aceh	21,8 rb
8.	Nusa Tenggara Barat	21,1 rb
9.	Kalimantan Timur	20,2 rb
10.	Riau	19,6 rb

West Java recorded the highest total transactions at **198.7 rb**, significantly surpassing all other provinces and indicating strong market activity in the region. Sumatera Utara (**48.2 rb**) and Jawa Tengah (**46.5 rb**) followed in second and third place, showing solid transaction volumes but with a considerable gap compared to West Java. Jawa Timur (**34.8 rb**) and Sulawesi Utara (**33.3 rb**) demonstrated moderate performance, while the remaining provinces—Sumatera Barat, Aceh, Nusa Tenggara Barat, Kalimantan Timur, and Riau—contributed lower yet consistent transaction levels; overall, the distribution reflects a **high concentration of transaction activity in West Java** relative to other provinces.

Data Analysis

High Branch Ratings, Stagnant Transaction Scores

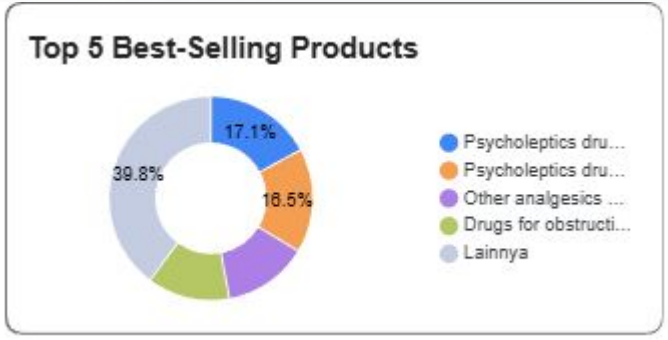
Top 5 Branches: Highest Rating, Lowe...

City	Branch ...	Transaction Ra...
Sorong	4,64	4
Cikampek	4,62	4
Kupang	4,6	4
Samarinda	4,6	4
Pematangsi antar	4,59	4

Branches such as Sorong, Cikampek, and Kupang demonstrate **strong overall performance** with branch ratings above **4.5**, reflecting **positive customer perception** at the outlet level. However, transaction ratings remain consistently at **4**, showing **no significant improvement**. This gap suggests that while customers value the **overall branch experience**, the **transaction process itself may not deliver exceptional value**. The stagnation in transaction ratings indicates **operational consistency without differentiation**, where service meets expectations but does not exceed them, limiting the potential for **higher customer satisfaction and loyalty**.

Data Analysis

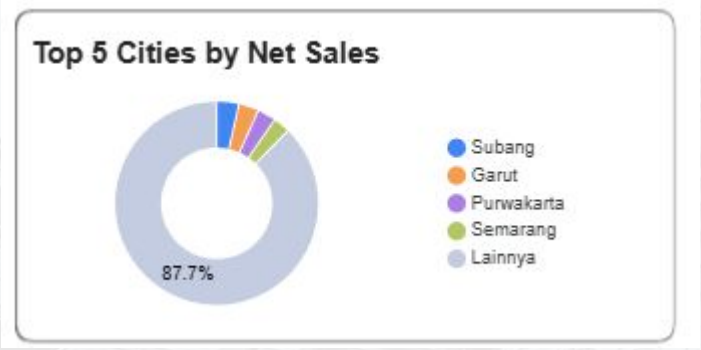
Diversified Product Sales Distribution



The top-selling products are dominated by the **"Others" category at 39.8%**, showing a **broad distribution of sales across multiple products**. Among the main categories, **Psycholeptics (17.1%)** and **Psychoanaleptics (16.5%)** contribute the largest shares. Overall, the results indicate a **diversified product portfolio**, with no single category overwhelmingly dominating sales.

Data Analysis

Sales Concentrated in West Java's Major Cities



The net sales distribution by city shows that **"Others" dominate with 87.7%**, indicating that revenue is heavily spread across cities outside the top five. Among the listed cities, Subang, Garut, Purwakarta, and Semarang contribute relatively small portions individually. Overall, the data reflects a **highly dispersed sales distribution**, with no single city significantly dominating total net sales.

Data Visualization

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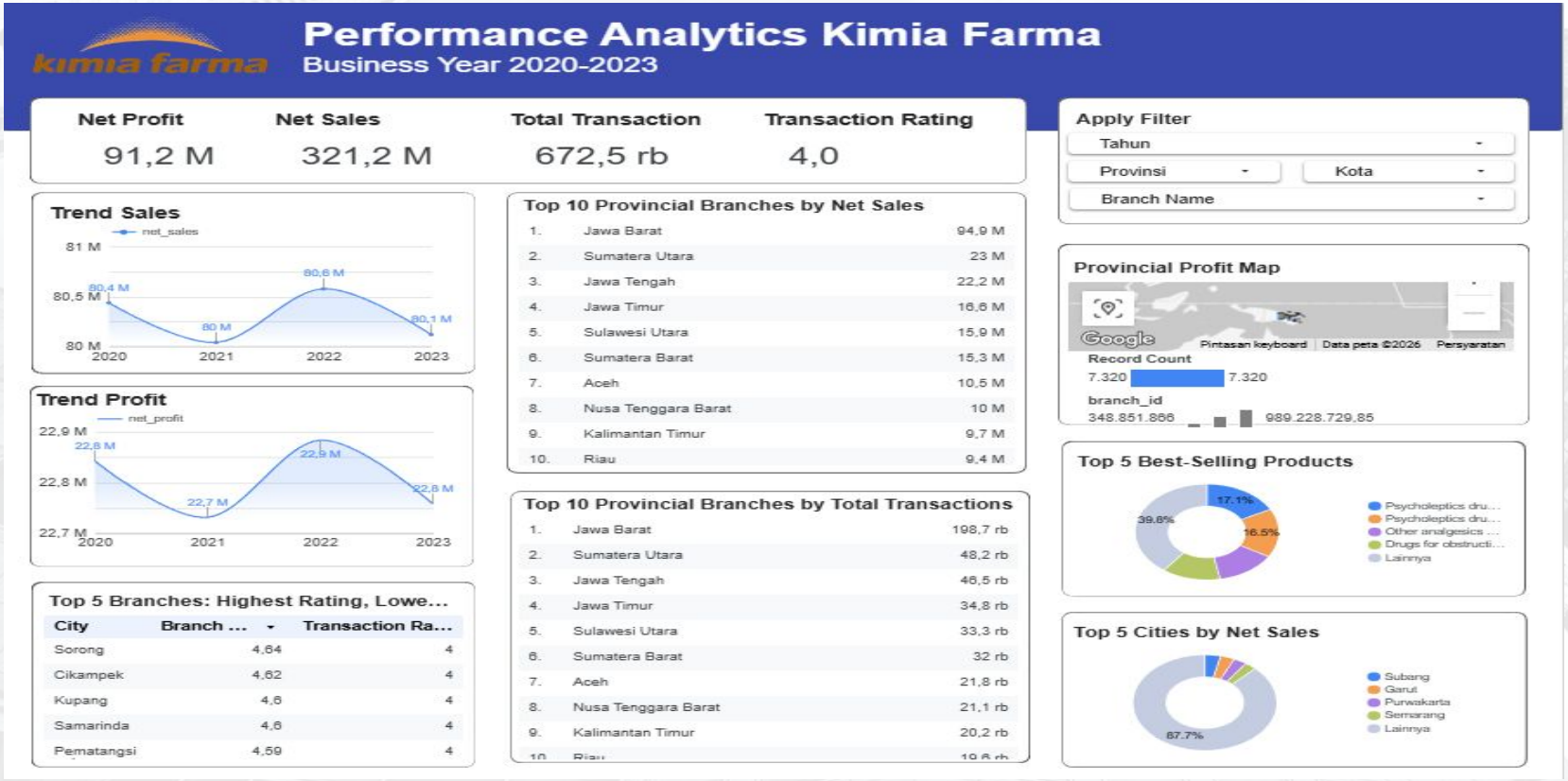
Dashboard



Insight

1. **Stable but Fluctuating Financial Performance**

Sales and profit remained relatively stable from 2020–2023, with 2022 marking the strongest recovery year after the 2021 slowdown.

2. **High Regional Profit Concentration**

Approximately 30% of total profit comes from Jawa Barat alone, and the top five provinces contribute around 55–60%, indicating strong regional imbalance.

3. **Transaction Activity Heavily Centered in West Java**

West Java leads significantly in total transactions, creating dependency on one primary market.

4. **Strong Branch Ratings but Transaction Experience Plateau**

Several branches have ratings above 4.5, yet transaction ratings remain at 4, suggesting operational consistency but limited service differentiation.

5. **Diversified Product Portfolio**

No single product category dominates sales, with “Others” contributing the largest share (39.8%), indicating a broad and diversified demand base.

Recommendations

1. **Reduce Regional Dependency**

Develop targeted expansion and performance improvement programs in mid-tier provinces to reduce over-reliance on Jawa Barat.

2. **Enhance Transaction Experience**

Improve checkout efficiency, digital payment integration, and customer service responsiveness to elevate transaction ratings beyond 4.

3. **Optimize High-Performing Regions**

Strengthen inventory management and promotional strategies in top-performing provinces to maximize profitability without oversaturation.

4. **Leverage Product Diversification**

Identify high-growth categories (e.g., Psycholeptics & Psychoanaleptics) and create focused marketing or bundling strategies to increase category contribution.

5. **Improve Underperforming Provinces**

Implement localized marketing campaigns, pricing adjustments, and operational audits to boost profitability in low-contribution regions.

Thank You

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